estimates were correct (and we knew they were highly uncertain), then the gravitational-wave signal would be *ten times smaller than Braginsky's standard quantum limit,* and therefore could not possibly be detected using a bar and any known kind of sensor.

Though this was extremely worrisome, all was not lost. Braginsky's deep intuition told him that, if experimenters were especially clever, they ought to be able to circumvent his standard quantum limit. There ought to be a new way to design a sensor, he argued, so that its unknowable and unavoidable kick does *not* hide the influence of the gravitational waves on the bar. To such a sensor Braginsky gave the name *quantum nondemolition*[3]; "quantum" because the sensor's kick is demanded by the laws of quantum mechanics, "nondemolition" because the sensor would be so configured that the kick would not demolish the thing you are trying to measure, the influence of the waves on the bar. Braginsky did not have a workable design for a quantum nondemolition sensor, but his intuition told him that such a sensor should be possible.

This time I listened, carefully; and over the next two years I and my group at Caltech and Braginsky and his group in Moscow both struggled, on and off, to devise a quantum nondemolition sensor.

We both found the answer simultaneously in the autumn of 1977—but by very different routes. I remember vividly my excitement when the idea occurred to Carlton Caves and me[4] in an intense discussion over lunch at the Greasy (Caltech's student cafeteria). And I recall the bittersweet taste of learning that Braginsky, Yuri Vorontsov, and Farhid Khalili had had a significant piece of the same idea in Moscow at essentially the same time—bitter because I get great satisfaction from being the first to discover something new; sweet because I am so fond of Braginsky and thus get pleasure from sharing discoveries with him.

Our full quantum nondemolition idea is rather abstract and permits a wide variety of sensor designs for circumventing Braginsky's standard quantum limit. The idea's abstractness, however, makes it difficult to explain, so here I shall describe just one (not very practical) example of a quantum nondemolition sensor.[5] This example has been called, by Braginsky, a *stroboscopic sensor.*

A stroboscopic sensor relies on a special property of a bar's vibrations: If the bar is given a very sharp, unknown kick, its amplitude of vibration